

JADE SATYKO HATANAKA MARQUES

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Work Experience

Machine Learning Engineer

Nov 2025 – current

Nubank, Full Time

- Trained, evaluated and optimized credit risk causal models in production; launched A/B experiments and built the models' monitoring infrastructure (ETL, dashboards, alerts) to track performance post-deployment.
- Drove the technical evaluation informing a new external credit bureau vendor purchase decision; trained and benchmarked credit limit elasticity models with and without the new signal via controlled ablation experiments to quantify predictive lift.
- Technologies: **Python, Scala, Databricks, SQL, Claude Code**

Machine Learning Engineer

Mar 2025 – Sept 2025

Amadeus, Internship

- Trained and deployed an Information Retrieval system that ranks past similar records for a new given incident.
- Trained, evaluated and deployed Multilabel Classifiers (BERT, Random Forest, XGBoost) that predict recovery actions for a new incident.
- Deployed an incident summarization tool (BART and OpenAI) to production via FastAPI on **Azure**.
- Technologies: **Python, Pandas, scikit-learn, PyTorch, HuggingFace, FastAPI, Azure, mlflow, Docker**

Machine Learning Engineer

June 2024 – current

VTEX, Research Assistant

- Conceived a LLM-based Product Attribute Value Extraction framework.
- Generated Knowledge Graphs to enhance Product Catalog and Chatbot assistants (RAG).
- Developed a LLM-as-a-judge framework using open-source LLMs to evaluate e-commerce search engine variations.
- Technologies: **Python, Pandas, LangChain, HuggingFace, ollama, openai, scikit-learn, AWS**

Automation/Data Engineer

June 2024 – Sept 2024

Moody's, Internship

- Developed a tool to automatically compare complex PostgreSQL databases (340+ tables across multiple schemas) and generate customized HTML reports highlighting their differences.
- Automated SQL query generation from specification spreadsheets by computing table relationships, reducing analysis time by 40% and manual workload significantly.
- Technologies: **Python, Pandas, SQL, Jenkins, Docker, AWS, Networkx, HTML, CSS**

Software Engineer

Jan 2023 – July 2023

Nubank, Internship

- Implemented algorithms that prevent fraudulent transactions (backend of the Frauds microservice), using Clojure.

Software Engineer

Nov 2022 – Jan 2023

SAP, Internship

- Contributed to the backend of SAP's Data Intelligence software, employing Go, Python, Docker and CI/CD techniques.

Machine Learning Engineer

Sept 2021 – Feb 2022

Universidade Federal do Rio Grande do Sul, Research Assistant

- Crafted Graph Neural Networks for social network analysis and performed data cleaning on a Psychology dataset.
- Technologies: **Python, Pandas, Pytorch**

Data Analyst and Pilot Cadet

Jan 2018 – Oct 2020

Brazilian Air Force Academy

- Designed and deployed a real-time data collection and analytics system, migrating from manual paper analysis to cloud. Reduced analysis time by approximately 60% through interactive dashboards.

Technical Skills

Data Ingestion and Processing: Pandas, NumPy, SQL, Spark, NetworkX, Matplotlib, Seaborn, MATLAB

Model Training and Tuning: PyTorch, TensorFlow, scikit-learn, xgboost

Model Deployment and MLOps: FastAPI, REST APIs, Docker, **Azure, AWS**

Monitoring and Retraining: MLflow, GitLab CI/CD, Jenkins

LLMs and RAG: HuggingFace Transformers, LangChain, chromaDB, SpaCy, NLTK, BERT

Other Engineering Skills: Git, Java, C, JavaScript, Bash, Go, Clojure, HTML, CSS, PostgreSQL, OracleDB, Agile

Languages

- English: C1 (TOEIC: 965/990)
- French: B2
- Portuguese: native
- Spanish: B1

Education

Masters in <u>Data Science and Artificial Intelligence</u> , graduated with honors <i>École Nationale Supérieure d'Informatique et de Mathématiques Appliquées de Grenoble (ENSIMAG)</i>	2023 - 2025 <i>*triple degree</i>
Information Systems Engineering , graduated with honors <i>École Nationale Supérieure d'Informatique et de Mathématiques Appliquées de Grenoble (ENSIMAG)</i>	2023 - 2025 <i>*triple degree</i>
B.Sc. in <u>Computer Science</u> , graduated with honors <i>Universidade Federal do Rio Grande do Sul (UFRGS)</i> <i>*triple degree: an agreement between ENSIMAG and UFRGS granted me 3 diplomas in 5 years</i>	2021 - 2026 <i>*triple degree</i>
Bachelor's in <u>Aeronautical Sciences</u> and bachelor's in <u>Public Management</u> <i>Brazilian Air Force Academy (AFA)</i>	2018 - 2020 <i>completed 3 out of 4 years</i>

Projects

BrainIAC Epilepsy Probing — Confound-Aware Linear Probing Study <i>Technologies: Python, PyTorch, scikit-learn, MLflow, Foundation Models</i> - Investigated whether BrainIAC, a SimCLR-pretrained brain MRI foundation model, linearly encodes epilepsy-relevant concepts via frozen CLS-token embeddings for 340 T1w/FLAIR scans from the Bonn FCD-II dataset. - Designed a confound-aware evaluation protocol: ran a scanner/era probe as an explicit baseline (AUC <u>0.986</u>), then built an <u>era-stratified re-analysis</u> with a <u>permutation null</u> to isolate genuine signal from confounds. - Trained <u>logistic-regression probes</u> with subject-level 5-fold cross-validation to detect focal cortical dysplasia (FCD-II) lesions and hemispheric lateralization directly from frozen embeddings, without any epilepsy-specific fine-tuning.	Research Project
DataSUS-RS — Public Health Analytics Platform <i>Technologies: Python, PySpark, DuckDB, Kafka, LightGBM, SHAP, Streamlit, Postgres, Docker</i> - Ingests 1.5M+ hospital records/year from 6 DataSUS sources (SIH, CNES, SIM, SIA, SINASC, SINAN) through a reproducible <u>Medallion pipeline</u> (Bronze → Silver → Gold) with PySpark and Parquet. - Built a real-time triage decision engine on Kafka using the Manchester protocol, scoring incoming patients with a <u>LightGBM mortality risk model</u> (AUC 0.879) and persisting alerts to PostgreSQL. - Delivered 13 interactive Streamlit dashboards, including hospital mortality maps, CID continuity analysis, and ML explainability via SHAP, surfacing epidemiological gaps across Rio Grande do Sul (Brazil).	Academic Project
FindshipAI Opportunity Recommendation System <i>Technologies: Python, Pandas, FastAPI, LLMs, Langchain, ChromaDB, AWS</i> Receives free-text input about user preferences and <u>ranks most relevant</u> scholarships/exchange programs/seasonal jobs.	Freelance project

Honours and Awards

- **SBC Outstanding Student Award** (Prêmio Aluno Destaque), 2026 — top academic performance in my cohort.
- **Outstanding Student Award** (Meninas Digitais), 2026 — academic excellence in the Brazilian CS community.
- **Outstanding Mentor**, Technovation Summer School for Girls, 2026 — students elected me as the most helpful mentor.
- **Finalist**, Microsoft/EY Women in GenAI 2024 hackathon — chatbot for Assurance Maladie agents using Generative AI.
- **Initiative d'Excellence Scholarship**, 2023, 2024 — for academic performance and extracurricular projects.
- **Palantir Women in Technology Scholarship - Global**, 2023 — one of 10 recipients globally.
- **DAC Young Fellows Best Paper Analysis Award and Scholarship**, 2022 — research on hypothetical processors.
- **Military and Intellectual Merit of the Brazilian Air Force Academy**, 2018, 2019, 2020 — all years I served

Volunteer Experience

Mentor at Technovation Summer School for Girls Mentored a team of 4 students through ideation to the development of an edtech app connecting university students for corporate-sponsored peer mentoring to reduce dropout rates.	Feb 2026 - Apr 2026
Coordinator at Program.ada Hosted lectures and Programming Workshops for low-income Brazilian Schools.	Jan 2023 - Aug 2026
Python Instructor at SAP's Digiport Taught Python and Programming Logic to Brazilian women who pursued developer roles at SAP.	Aug 2022 - Sept 2022
Product Manager at SAP Tech School Organized classes on Technology for low-income Brazilian High School students.	Jul 2022 - Jan 2023
Mathematics Instructor at PreparaEng (UFRGS) Reinforced Mathematics concepts to Engineering students, aiming to reduce the dropout rate from Engineering courses.	Sept 2021 - Dec 2021